

PH 613: Quantum Mechanics II

Course Code	PH 613
Course Name	Quantum Mechanics II
L-T-P-C	3-0-0-3
Prerequisites	First course on Quantum Mechanics
Intended for	Core for M.Sc. and I-Ph.D. in Physics; Elective for others
Mutual Exclusion	None
Approval	10th Senate (Jun 2016)
Revised	64th BoA (May 2026) - syllabus revised

Preamble

This course introduces some of the advanced level topics in quantum mechanics. The course begins a review of some of the basic concepts in quantum mechanics and then discusses the detailed theory of angular momentum algebra. It will then proceed to discuss the concepts in scattering theory.

Course Modules

- **Unit/Topic 1:** Review of basic concepts in quantum mechanics, measurements, observables and generalized uncertainty relations, change of basis, generator of translation. (8 hours)
- **Unit/Topic 2: Angular Momentum:** General theory of angular momentum, Angular momentum algebra, Addition of angular momenta, Clebsch-Gordon coefficients, illustration of Wigner-Eckart theorem. (8 hours)
- **Unit/Topic 3: Symmetries in Quantum Mechanics:** Symmetry principles in quantum mechanics, conservation laws and degeneracies, discrete symmetries, parity and time reversal. (8 hours)
- **Unit/Topic 4:** Oscillating dipoles and Bloch sphere representation, Density Operator and Density Matrix formalism, Optical Bloch Equations (8 hours)
- **Unit/Topic 5: Scattering Theory:** Non-relativistic scattering theory. Scattering amplitude and cross-section, Partial wave analysis, optical theorem, Integral equation for scattering, Born approximation. (10 hours)

Laboratory/practical/tutorial Modules: No

Text Books

1. *Modern Quantum Mechanics* – J J Sakurai (Addison Wesley, revised edition, 1993).
2. *Quantum Mechanics*, Cohen-Tannoudji, B Diu, F Laloe (Vol. I and II) (Wiley, second edition 2020)

References

1. *Many-electron theory* – S. Raimes (North-Holland Pub. Co. 1972)
2. *Quantum Mechanics* – P. C. Deshmukh (Cambridge University Press 2023)
3. *Quantum Mechanics* – Merzbacher (Third edition, Wiley, 2011)
4. *Quantum mechanics* – Landau and Lifshitz (Butterworth-Heinemann Ltd; 3rd Revised edition (18 December 1981))